

Bone mechanics. Solutions.

1. Suppose that the temporal activity of the space-clamped nerve axon is modelled with the two variable FitzHugh-Nagumo model with an external applied current I_a :

$$\frac{dv}{dt} = f(v) - w + I_a, \quad \frac{dw}{dt} = bv - w,$$

where the function $f(v)$ is the piecewise linear function:

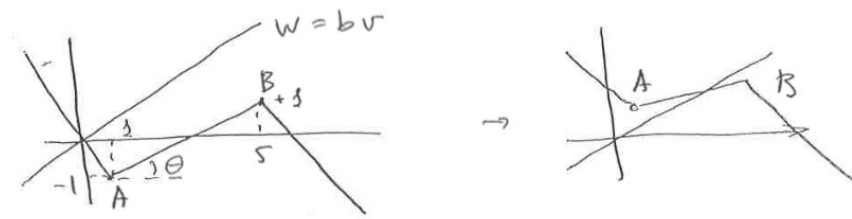
$$f(v) = \begin{cases} -v, & \text{for } v \leq 1 \\ \frac{1}{2}v - \frac{3}{2}, & \text{for } 1 \leq v \leq 5 \\ 6 - v, & \text{for } v \geq 5 \end{cases}$$

and I_a is a positive constant.

Find the range of values of the parameters b and I_a for which the model has a periodic behaviour.

Solution

Let points A and B have coordinates $A(v_1, w_1)$ and $B(v_2, w_2)$. Then $\tan \Theta = \frac{w_2 - w_1}{v_2 - v_1}$. For a periodic behaviour we need i) $\tan \Theta < b$ and ii) point A lies above bv and point B lies below bv .



$$\tan \Theta = 2/4 = 1/2 < b, \text{ so } b > 1/2$$

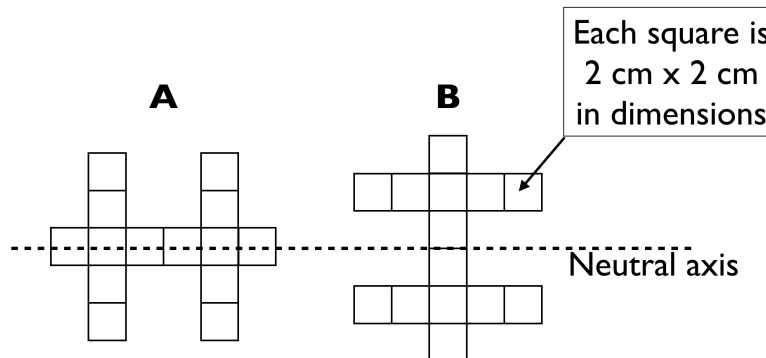
and $-1 + I_a > b$, $I_a > b + 1$ and $1 + I_a < 5b$, $I_a < 5b - 1$

So

$$b > \frac{1}{2}, \text{ and } b + 1 < I_a < 5b - 1$$

to have periodic solution.

2. Consider two beams A and B of equal cross-sectional area, with the material distributed around the neutral axis/plane as shown in the figure below. Note that the area of the square used to construct the shapes is equal to $2\text{cm} \times 2\text{cm} = 4\text{cm}^2$. Calculate the area moment of inertia I_A for these two cases. The beam of which shape, A or B , will resist the bending better? Why?



Solution

For the shape A

$$I_A = 4 \int_{-1}^1 y^2 * 2 \, dy + 2 \int_{-5}^5 y^2 * 2 \, dy = 338.6$$

For the shape B

$$I_A = \int_{-6}^6 y^2 * 2 \, dy + 4 \int_{-2}^2 y^2 * 4 \, dy = 586.6$$

The beam B will resist bending better because its moment of inertia is larger (more material is located more distant from the neutral axis).